

due to a delay in releasing the order to the market or due to inferior performance by the trader or by the algorithm. Knowing who is responsible for cost will help investors improve the process of lowering transaction costs going forward. Wagner's expanded implementation shortfall categorizes cost into delay, trading, and opportunity related cost. Perold's original formulation did not separate delay and trading related costs when they occurred during the implementation phase. Wagner's formulation of implementation shortfall is what makes it possible to measure performance across traders, brokers, and algorithms.

The derivation of the expanded implementation shortfall is as follows.

First, define two time horizons: investment and trading. The investment horizon is the time from the investment decision t_d to beginning of trading t_0 . The trading horizon is the time from beginning of trading t_0 to the end of trading t_n . The corresponding prices at these time intervals are P_d , which is the decision price, P_0 which is the price at beginning of trading, also known as the arrival price, and P_n , which is the price at the end of trading. All prices are taken as the mid-point of the bid-ask spread if during market hours or the last traded price or official close if after hours.

Next, rewrite the price change over these two intervals as follows:

$$(P_n - P_d) = (P_n - P_0) + (P_0 - P_d)$$

Now substitute this price into Perold's implementation shortfall:

$$IS = \left(\sum s_j p_j - \sum s_j P_d \right) + \left(S - \sum s_j \right) \cdot (P_n - P_d) + fees$$

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This expression can then be written based on our investment and trading horizons and is known as the Expanded Implementation Shortfall or Wagner's Implementation Shortfall. This is as follows:

$$\text{Expanded IS} = \underbrace{S(P_0 - P_d)}_{\text{Delay Related}} + \underbrace{\sum s_j p_j - \left(\sum s_j \right) P_0}_{\text{Trading Related}} + \underbrace{\left(S - \sum s_j \right) (P_n - P_0)}_{\text{Opportunity Cost}} + fees$$